Interpretable Survival Modeling for Lung Adenocarcinoma via GAM-Based Meta-Learning

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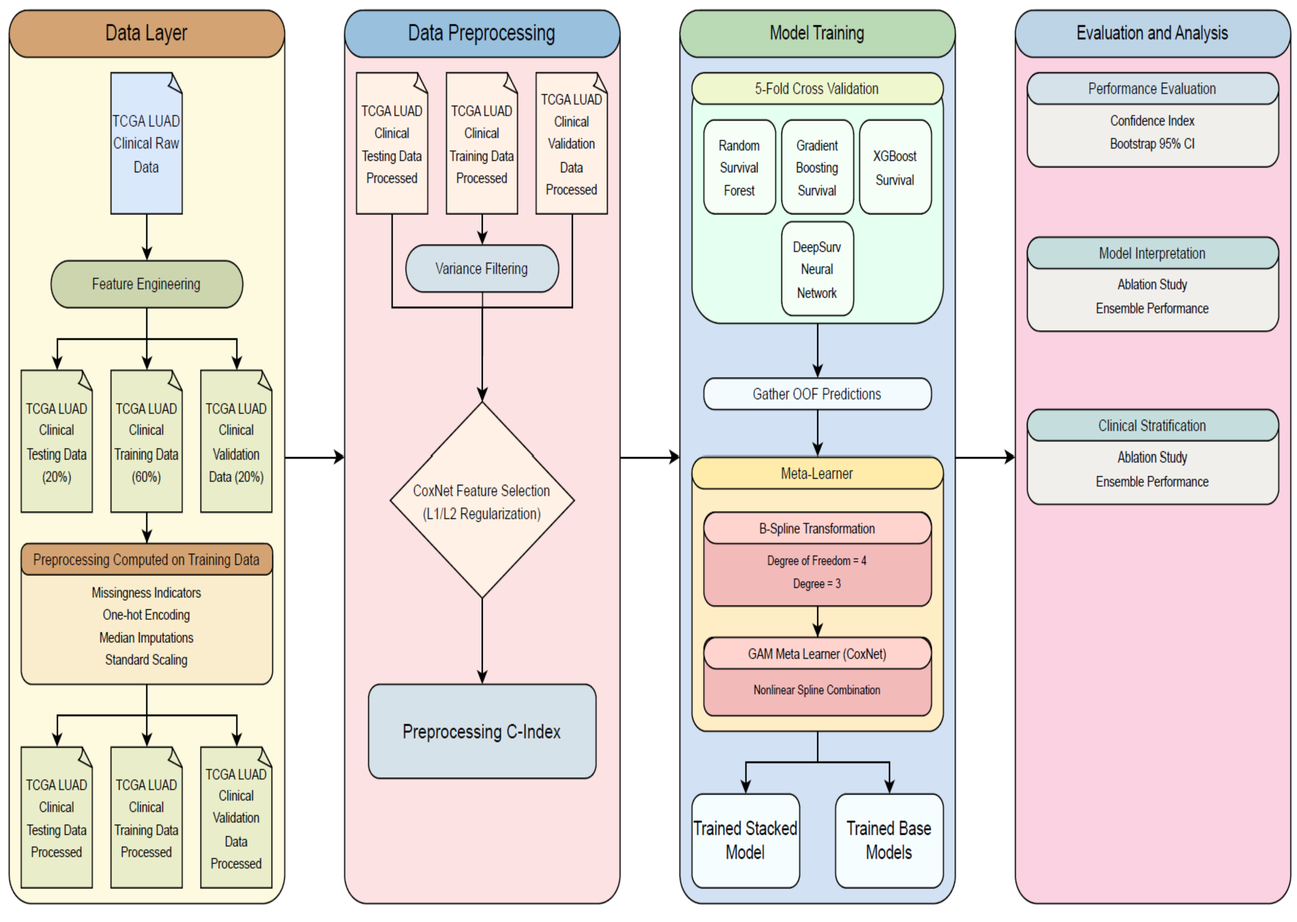
ABSTRACT

Survival prediction for lung adenocarcinoma (LUAD) remains challenging due to heavy censoring in clinical datasets. To address data sparsity, ensemble machine learning has shown promise, while most deep learning models sacrifice interpretability and risk overfitting or unstable performance. This work aims to provide an interpretable "glass-box" framework that achieves both high interpretability and improved survival prediction in patients with LUAD. To select the most discriminative subset of features, penalized CoxNet was applied to LUAD cases from the Cancer Genome Atlas (TCGA). The resulting feature subset was then used to train four survival models: Random Survival Forest (RSF), Gradient Boosting Survival Analysis (GBS), Extreme Gradient Boosting (XGB), and DeepSurv Neural Network. The model predictions from out-of-fold samples were combined using a generalized additive model (GAM) meta-learner, which provides transparent weighting for the contribution of each model to the ensemble. Cross-validated ablation analysis demonstrated that RSF, GBS, and DeepSurv consistently contributed positively to ensemble performance, while XGB provided no measurable benefit. Based on these findings, we constructed an optimized three-model ensemble (RSF + GBS + DeepSurv). This model achieved a concordance index (C-index) of 0.671 (64% DeepSurv, 21% GBS, and 15% RSF), outperforming the four individual models while remaining fully interpretable. Kaplan–Meier analysis was used to evaluate model-derived risk stratification, demonstrating strong clinical stratification, with high-risk patients exhibiting significantly shorter survival compared to the low-risk group (log-rank $p = 3.06 \times 10^{-3}$). Although the medium-risk group did not reach median survival during follow-up, survival curves showed a clear ordinal separation between risk strata. These results demonstrate that combining biologically grounded feature selection with interpretable ensemble learning yields improved predictive accuracy and clinically meaningful risk stratification for LUAD, supporting the use of AI models in precision oncology.

METHODOLOGY

Architecture Pipeline



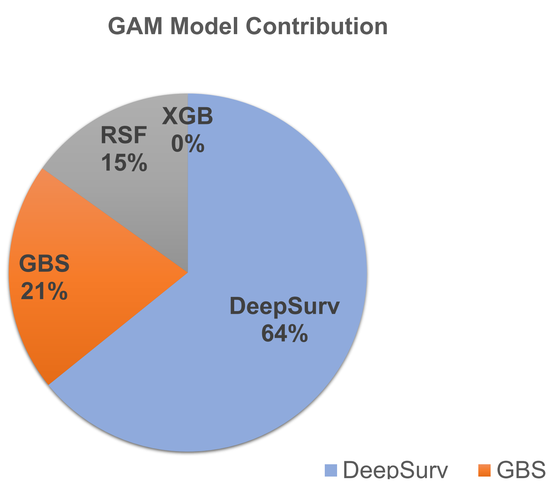
RESULTS

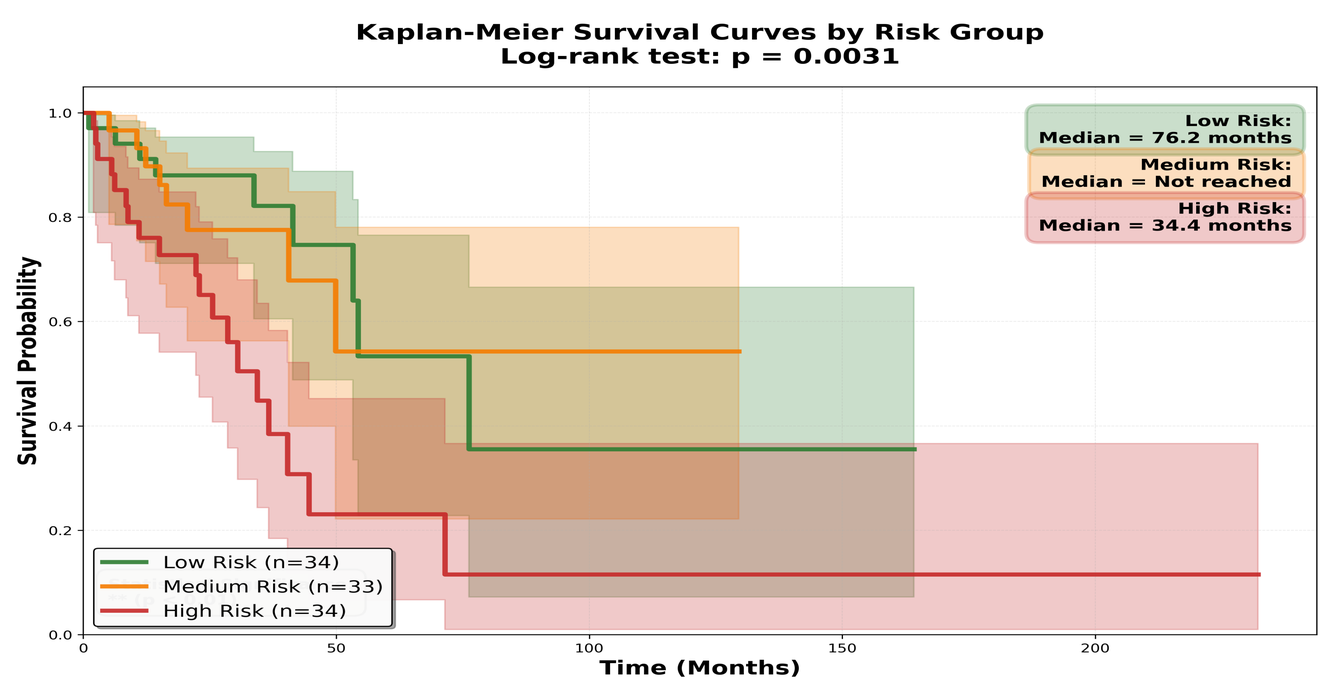
GAM Ensemble C-Index 0.671
95% Confidence Interval : [0.533 – 0.772]
Log Rank p = 0.003 [Highly Significant]

- TCGA-LUAD: 505 patients :
  - 182 events (36%)
  - Censored 323 (64%)
- CoxNet Feature Selection (L1=0.9): 25 features selected
- Base Models (5-fold CV): RSF GBS XGB DeepSurv
- GAM Meta-Learner: B-splines (df=4) with CoxNet ensemble (L1=0.9)
- Test Performance: C-index 0.671 [95% CI: 0.533-0.772], Log-rank p=0.003

Model Performance Comparisons

MODEL	Train CV	Validation	Test
RSF	0.656	0.611	0.656
GBS	0.623	0.628	0.637
XGB	0.629	0.605	0.617
DeepSurv	0.682	0.644	0.643
GAM Ensemble	0.684	0.649	0.671





CONCLUSION

This study demonstrates that the GAM-based meta-learner combining multiple survival models (RSF, GBS, DeepSurv) achieves robust predictive performance (C –index = 0.671) while providing interpretable ensemble-level contributions through B-spline transformations. The purposed framework successfully stratifies lung adenocarcinoma patients into distinct risk groups and supports the integration of transparent ensemble methods into precision oncology, offering explainability relevant to clinical decision-making. Future work will focus on validation in independent cohorts and the incorporation of additional data modalities to further improve risk stratification.

FUTURE DIRECTIONS

- Validations on independent external cohorts
- Perform feature Ablation
- Spline complexity optimization
- Alternative meta-learner comparisons (e.g., stacking, voting)
- Extension of framework to other cancer type